Vectors Physics IIT JEE Notes PDF

Vectors Physics IIT JEE notes PDF includes all-important formulas, theorems, facts, and equations of vectors. Vectors Physics IIT JEE notes PDF is beneficial for students who are preparing for the JEE exam. Referring to these notes will help students to have a complete revision before the exam. It also helps students to memorize all the important facts about vectors.

In Physics, we classify quantities into vectors and scalars. The quantities which have both magnitude and direction are called vectors. Examples are velocity, force, displacement, weight, acceleration, etc. The quantities which have only magnitude and no direction are called scalar quantities. Examples are mass, volume, speed, time, frequency, etc.

A quantity is said to be a vector if it satisfies the following conditions.

(a) Obeys the law of parallelogram addition.

(b) Has a specified direction.

(c) The addition is commutative. i.e A+B = B+A

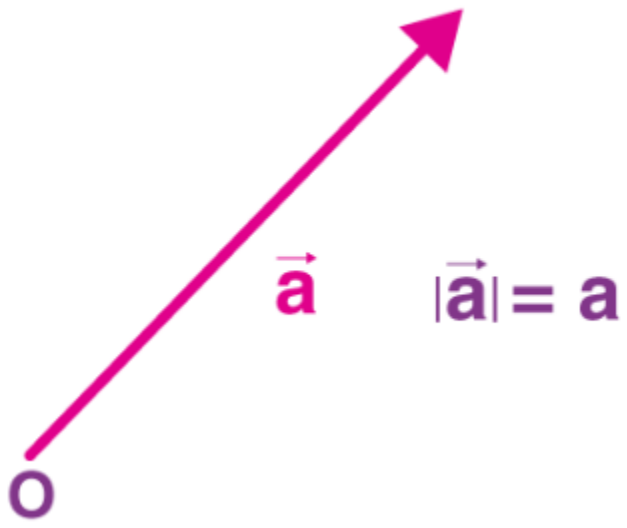
Table of Contents

- Representation of a Vector
- Position Vectors
- Dot Product of Two vectors
- Cross Product of Two Vectors
- Projection of a Vector
- Solved Examples
- Practice Problems
- Frequently Asked Questions

Representation of a Vector

A vector is represented by a line with an arrowhead. The point O from which the arrow starts is called the tail or initial point or origin of the vector. Point A where the arrow ends is called the tip or head or terminal point of the vector. A vector displaced parallel to itself remains unchanged. If a vector is rotated through an angle other than 360^0, it changes.

A vector can be replaced by another when its direction and magnitude are the same.



Unit Vector

A vector having unit magnitude is called a unit vector. It is used to denote the direction of a given vector.

$$\vec{A} = \hat{a}A$$

$\hat{a}$ is the unit vector along the direction of $\vec{A}$

Types of Vectors

(i) Negative of a Vector: It has the same magnitude but opposite direction of the given vector.

(ii) Equal vectors: If two vectors have equal magnitude and the same direction, then they are equal vectors.

(iii) Collinear vectors: Two vectors acting along the same straight lines or along parallel straight lines in the same direction or in the opposite direction are called collinear vectors.

(iv) Coplanar vectors: If three or more vectors lie in the same plane, then they are called coplanar vectors.

(v) Zero vector: It is a vector with zero magnitude and no specific direction.

Addition of Vectors

Law of Triangle: If two sides of a triangle are shown by two continuous vectors (vector A and vector B), then the third side of the triangle in the opposite direction shows the resultant of two vectors. (vector C).

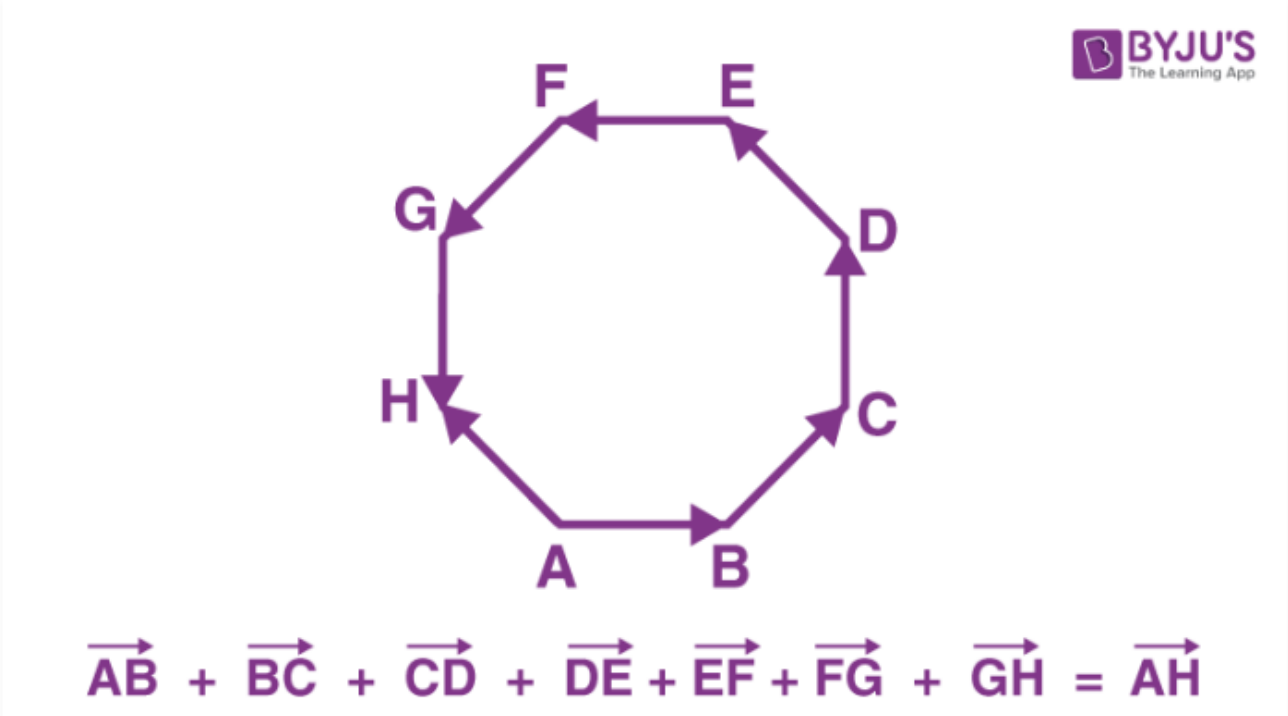
$$\vec{C} = \vec{A} + \vec{B}$$

Vector addition is commutative. => A + B = B + A

Vector addition is associative. => A+(B+C) = (A+B)+C

If all sides of a polygon are represented by continuous vectors, the vector sum of all sides is zero.

Polygon method: We use this method when we have to add more than two vectors. It is an extension of the triangular law of vector addition. If a number of vectors can be represented in magnitude and direction by the sides of a polygon taken in the same order, then their resultant is represented in magnitude and direction by the closing side of the polygon taken in the opposite order.



Subtraction of vectors

While subtracting vectors, we change the direction of the vector to be subtracted and then add.

$$\vec{A} - \vec{B} = \vec{A} + (\overrightarrow{-B})$$

Null vector

If a vector A is multiplied by zero, we get a vector whose magnitude is zero called null vector or zero vector. The unit of vector does not change on being multiplied by a dimensionless scalar.

Properties of Null vector

1) It has an arbitrary direction.

2) It is represented by a point.

3) It has zero magnitude.

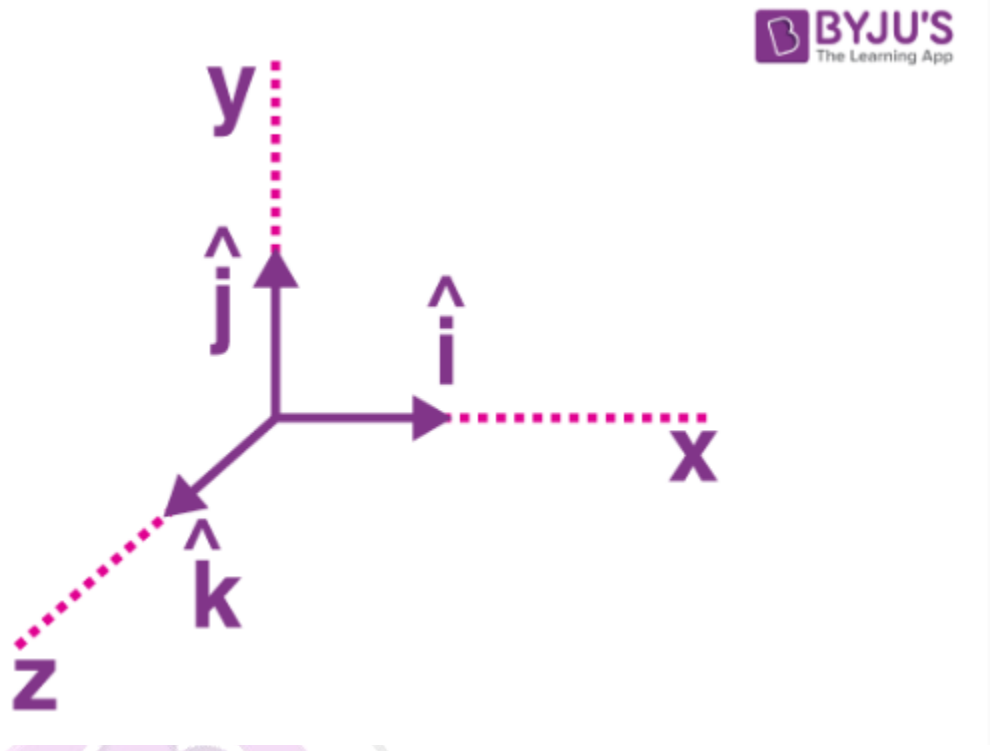
4) Dot product of a null vector with any vector is always zero.

5) Cross product of a null vector with any vector is also a null vector.

6) When a null vector is added or subtracted from a given vector, the resultant vector is the same as the given vector.

Orthogonal unit vectors

The unit vectors along the X-axis, Y-axis, and Z-axis of the right-handed cartesian coordinate system are written as $\hat{i}, \hat{j}\ and \hat{k}$ respectively. These are known as orthogonal unit vectors.



Components of a Vector

Consider a vector V. The components of a vector in a 2D coordinate system are considered to be x-component and y-component. We can represent V = (v_x, v_y). Let θ is the angle formed between the vector V and x-component of the vector. The vector V and its x-component (v_x) form a right-angled triangle if we draw a line parallel to the y-component (v_y).

The horizontal component $v_x = V \cos \theta$

Vertical component $v_y = V \sin \theta$

Position Vectors

Position vector is a vector that gives the position of a point with respect to the origin of the coordinate system. The magnitude of the position vector is the distance of the point P from the origin O. Vector OP is the position vector that gives the position of the particle with reference to O.

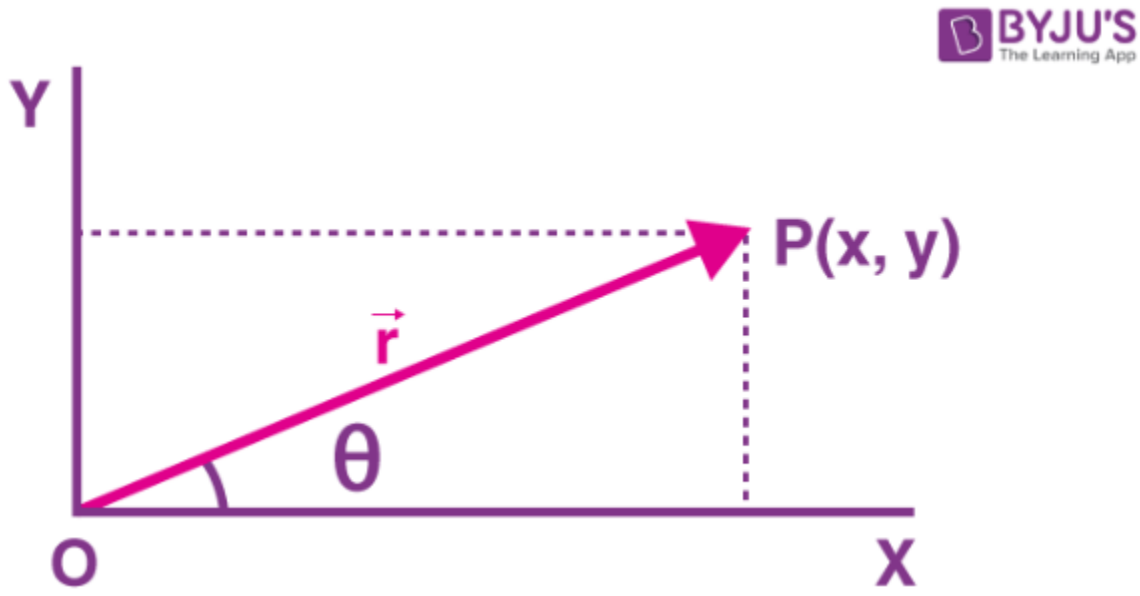
Consider point P whose coordinates are (x, y).

$\overrightarrow{OP} = \vec{r}$

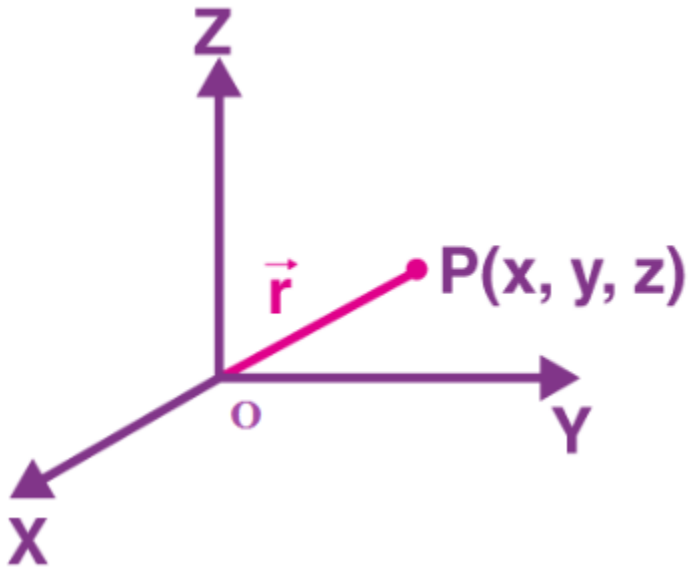
x = r cos θ, y = r sin θ

$\vec{r} = x\hat{i} + y\hat{j}$

$|\vec{r}| = \sqrt{x^2 + y^2}$



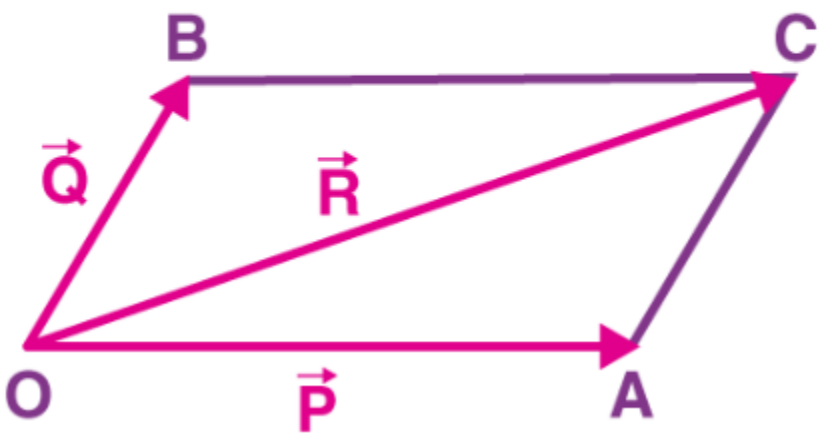
For a point P(x,y,z) in 3D coordinate system, $\overrightarrow{OP}$ or vector r is the position vector with origin O as the initial point.



Magnitude of vector OP = $\left|\overrightarrow{OP}\right| = \sqrt{x^2 + y^2 + z^2}$

Parallelogram Law of Vectors

If two vectors act along two adjacent sides of a parallelogram having a magnitude equal to the length of the sides, both pointing away from the common vertex, then the resultant is given by the diagonal of the parallelogram passing through the same common vertex, and in the same sense as the two vectors.



$$\vec{P} + \vec{Q} = \vec{R}$$

Or

$$\vec{OA} + \vec{OB} = \vec{OC}$$

Dot Product of Two Vectors

If vector A and vector B are two given vectors and θ is the angle between them, then

$$\vec{A}.\vec{B} = AB\cos\theta$$

A and B are the magnitude of vector A and vector B.

AB cos θ is a scalar quantity. B cos θ is the component of vector B in the direction of vector A.

The dot product of two vectors is the product of the magnitude of one vector with the resolved component of the other in the direction of the first vector.

The dot product is also called a scalar product.

Properties of Dot Product

1) Dot product of two vectors is commutative, i.e. a.b = b.a = ab cos θ.

2) The dot product of a vector to itself is the magnitude squared of the vector.

$$\vec{A}.\vec{A} = AA\cos 0 = A^2$$

3) The dot product of two mutually perpendicular vectors is zero.

$$\vec{A}.\vec{B} = AB\cos 90 = 0$$

$$\hat{i}.\hat{i} = 1$$

$$\hat{j}.\hat{j} = 1$$

$$\hat{k}.\hat{k} = 1$$

4) Dot products are distributive.

$$\vec{A}.(\vec{B} + \vec{C}) = \vec{A}.\vec{B} + \vec{A}.\vec{C}$$

5) The scalar product of two parallel vectors is equal to the product of their magnitudes.

$$\vec{A}.\vec{B} = AB\cos 0 = AB$$

Cross Product of Two Vectors

The cross product of two vectors, A and B, is denoted by A × B. Its resultant vector is perpendicular to A and B. Cross products are also called vector products. Cross product of two vectors will give the resultant a vector and calculated using the Right-hand Rule.

$$\vec{A} \times \vec{B} = AB \sin\theta \hat{n}$$

Consider two vectors A and B. The cross product of A and B is a vector having a magnitude equal to the product of the magnitudes of the two vectors and the sine of the angle between them, and having the direction perpendicular to the plane containing these vectors. θ is the angle between them and $\hat{n}$ is the unit vector perpendicular to the plane of vector A and vector B.

Properties of Cross Product

(a) Cross product is not commutative. Consider two vectors A and B.

A×B ≠ B×A

A×B = – B×A

(b) Cross product is distributive. Consider three vectors A, B, and C.

A×(B+C) = A×B + A×C

(c) The cross-product of two parallel vectors is a zero vector.

$$\vec{A} \times \vec{B} = AB \sin\theta \hat{n} = 0$$

For parallel vectors, θ = 0. So sin θ = 0. Thus

A×B = 0

(d) The cross-product of a vector by itself is a null vector.

A×A = 0

(e) The magnitude of the cross product of 2 vectors that are at right angles is equal to the product of the vectors.

If θ = 90^0

$$\vec{A} \times \vec{B} = AB \sin 90 \hat{n} = AB\,\hat{n}$$

(f) The cross product of unit orthogonal vectors i, j, k is as follows.

(i)

$\hat{i} \times \hat{j} = \hat{k}$

$\hat{j} \times \hat{k} = \hat{i}$

$\hat{k} \times \hat{i} = \hat{j}$

$\hat{j} \times \hat{i} = -\hat{k}$

$\hat{k} \times \hat{j} = -\hat{i}$

$\hat{i} \times \hat{k} = -\hat{j}$

(ii)

$\hat{i} \times \hat{i} = 0$

$\hat{j} \times \hat{j} = 0$

$\hat{k} \times \hat{k} = 0$

(g) The vector product can be expressed in terms of determinants.

Let

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Projection of a Vector

The projection of a vector $\vec{A}$ on other vector $\vec{B}$ is $\frac{\vec{A}.\vec{B}}{|\vec{B}|}$. If $\hat{i}$ is a unit vector along a line L, the projection of vector a on the line L is given by $\vec{a}.\hat{i}$. The projection vector of $\vec{AB}$ is zero, if $\theta = \pi/2$.

The projection vector of $\vec{AB}$ is $\overrightarrow{BA}$, if $\theta = \pi$.

Points to remember

(1) The sum of three non-coplanar forces cannot be zero.

(2) The minimum number of equal forces required for a zero resultant is two.

(3) The minimum number of unequal forces required for a zero resultant is three.

(4) Multiplication of velocity vectors by time gives the displacement.

(5) The unit of a vector is not changed if it is multiplied by a dimensionless scalar.

Question 1. Given A = 2i+3j and B = i+j. The component of vector A along vector B is

(a) $1/\sqrt{2}$

(b) $3/\sqrt{2}$

(c) $5/\sqrt{2}$

(d) $7/\sqrt{2}$

Solution:

Component of vector A along B is A.B/|B|

= $(2i+3j).(i+j)/\sqrt{(1+1)}$

= $5/\sqrt{2}$

Hence option c is the answer.

Question 2. A force is inclined at an angle 60^0 to the horizontal. If its rectangular component in the horizontal direction is 50 N, then the magnitude of the force in the vertical direction is

(a) 25 N

(b) 75 N

(c) 87 N

(d) 100 N

Solution:

Given horizontal component of force, f_x= 50 N

Given angle = 60^0

tan 60 = f_x/f_y

$\sqrt{3} = 50/f_y$

So $f_y = 50\sqrt{3}$

= 87 N

Hence option c is the answer.

Question 3. How many minimum numbers of coplanar vectors having different magnitudes can be added to give zero resultant?

(a) 2

(b) 3

(c) 4

(d) 5

Solution:

From the Triangle Law of vector addition, a minimum of three vectors is needed to get zero resultant. A minimum of 3 coplanar vectors is required to represent the same physical quantity having different magnitudes that can be added to give zero resultant.

Hence option b is the answer.

Question 4. The square of the resultant of two equal forces is three times their product. The angle between the forces is

(a) π

(b) $\pi/3$

(c) $\pi/4$

(d) $\pi/2$

Solution:

Let A and B be the two forces and θ be the angle between them.

Also A = B.

The square of the resultant of two equal forces is three times their product.

$F^2_{res} = 3AB$

=> $F^2_{res} = A^2+B^2+2AB \cos\theta$

=> $3AB = A^2+B^2+2AB \cos\theta$

Since A = B,

=> $3A^2 = 2A^2+2A^2\cos\theta$

=> $A^2 = 2A^2\cos\theta$

=> $\cos\theta = 1/2$

=> $\theta = \pi/3$

Hence option b is the answer.

Question 5. If A = B + C and the values of A, B, and C are 13, 12, and 5, respectively, then the angle between A and C will be

(a) $\cos^{-1}(5/13)$

(b) $\cos^{-1}(13/12)$

(c) $\pi/2$

(d) $\sin^{-1}(5/12)$

Solution:

Given A = B + C

Here $13^2 = 12^2 + 5^2$

So according to Pythagoras theorem, the angle between B and C is 90^0

Angle between A and C = $\cos\theta = 5/13$.

=> $\theta = \cos^{-1}(5/13)$

Hence option a is the answer.

Question 6: If the scalar and vector products of two vectors A and B are equal in magnitude, then the angle between the two vectors is

(a) 45^0

(b) 90^0

(c) 180^0

(d) 360^0

Solution:

Given A.B = A×B

=> $AB\cos\theta = AB\sin\theta$

=> $\tan\theta = 1$

=> $\theta = 45^0$

Hence option a is the answer.

Question 7: If the angle between vectors A and B is θ, then the value of the product (B×A).A is equal to

(a) $BA^2\cos\theta$

(b) $BA^2\sin\theta$

(c) zero

(d) $BA^2\sin\theta\cos\theta$

Solution:

B×A will be perpendicular to both A and B.

(B×A).A = (B×A)A $\cos\theta$ (here $\theta = 90^0$)

= |B×A||A| cos 90

= 0

Hence option c is the answer.

Practice Problems

1. If the angle between two vectors A and B is 120^0 then its resultant C will be

(a) C = |A-B|

(b) C < |A-B|

(c) C > |A-B|

(d) C = |A+B|

2. Three concurrent coplanar forces 1 N, 2 N, and 3 N acting along different directions on a body

(a) can keep the body in equilibrium if 2 N and 3 N act at right angles

(b) can keep the body in equilibrium if 1 N and 2 N act at right angles

(c) cannot keep the body in equilibrium

(d) can keep the body in equilibrium if 1 N and 32 N act at right angles

3. Vector A has a magnitude of 5 units and lies in the XY- plane, and points in a direction 120^0 from the direction of increasing X. Vector B has a magnitude of 9 units and points along the Z-axis. The magnitude of cross product A x B is

(a) 30

(b) 35

(c) 40

(d) 45

4. If a.b = |a×b|, then θ will be

(a) 45^0

(b) 60^0

(c) 30^0

(d) 75^0

5. The direction of A is vertically upward and the direction of B is in the North direction. The direction of A × B will be

(a) western direction

(b) eastern direction

(c) vertically downward

(d) at 45^0 upward north direction

6. A motorboat covers a given distance in 6 h moving downstream on a river. It covers the same distance in 10 h moving upstream. The time it takes to cover the same distance in still water is

(a) 9 h

(b) 7.5 h

(c) 6.5 h

(d) 8 h

Frequently Asked Questions

State the Polygon law of vector addition

The Polygon law of vector addition states that if a number of vectors can be represented in magnitude and direction by the sides of a polygon taken in the same order, then their resultant is represented in magnitude and direction by the closing side of the polygon taken in the opposite order.

Define vectors and scalars. Give two examples of each.

Vectors are quantities that have both magnitude and direction. Examples are displacement, weight, acceleration, etc. Scalars are quantities that have only magnitude and no direction. Examples are speed, time, frequency, etc.

What are equal vectors?

Vectors that have the same magnitude and the same direction are called equal vectors. These vectors may start at different positions.

State the parallelogram law of vector addition.

Parallelogram law of vector addition states that if two vectors act along two adjacent sides of a parallelogram having magnitude equal to the length of the sides, both pointing away from the common vertex, then the resultant is represented by the diagonal of the parallelogram passing through the same common vertex and in the same sense as the two vectors.

What are the different types of vectors?

The different vectors are unit vector, zero vector, equal vector, collinear vector, coplanar vector. position vector, co-initial vector, and like and unlike vectors.

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